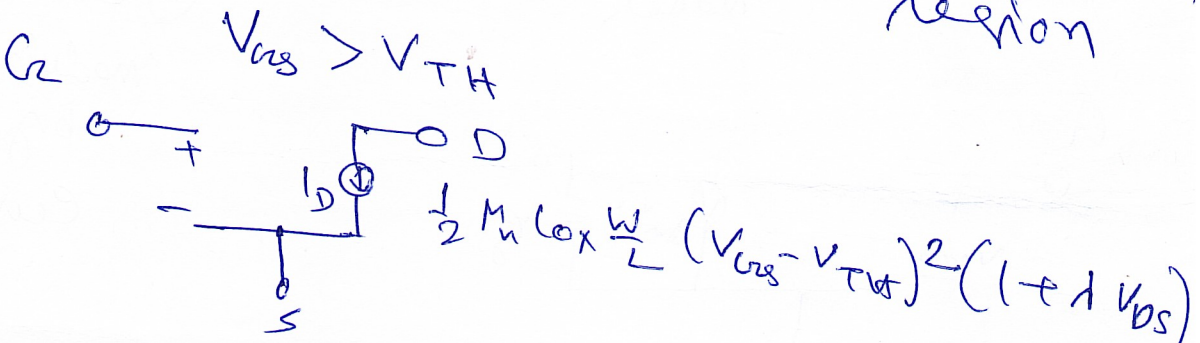
MOS Device Models

(a) Large-Signal Model

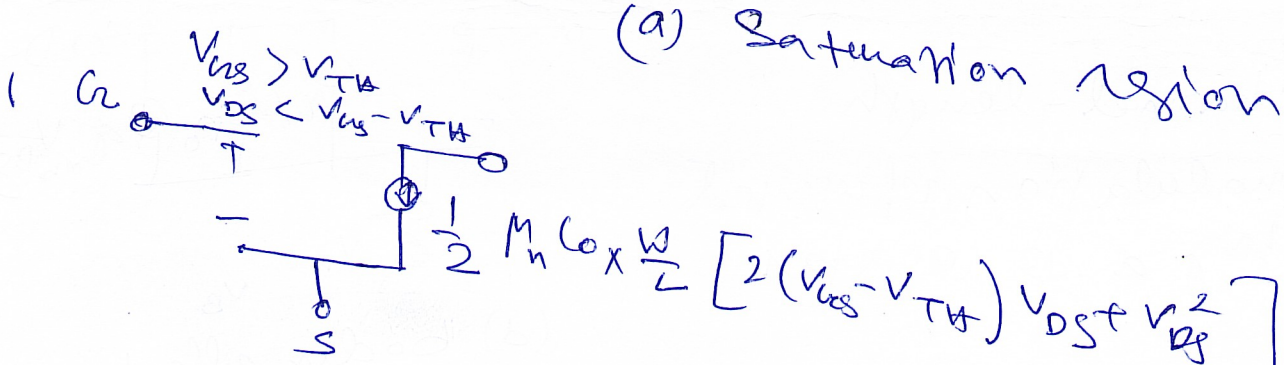
For an arbitrary voltage and current we can write

$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}\left[2(V_{GS}-V_{TH})V_{DS} - V_{DS}^2\right] \quad \text{Triode region}$$

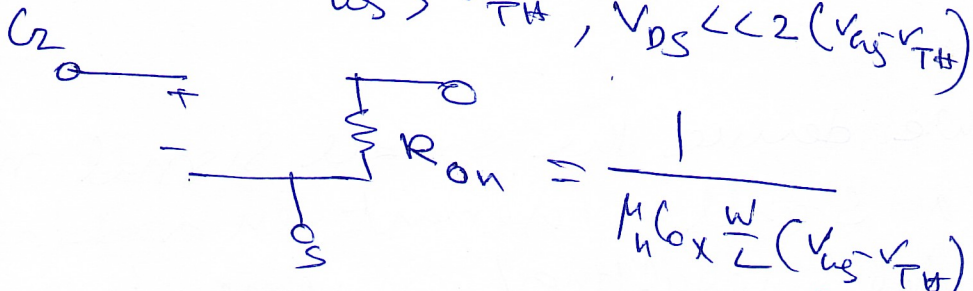
$$I_D = \frac{1}{2}\mu_n C_{ox}\frac{W}{L}(V_{GS}-V_{TH})^2(1+\lambda V_{DS}) \quad \text{Saturation region}$$



(a) Saturation region



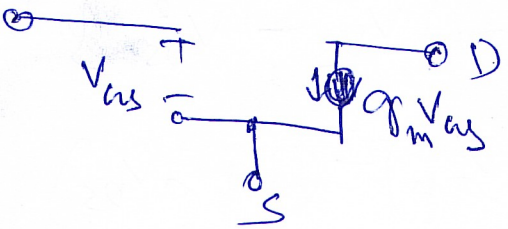
(b) triode region



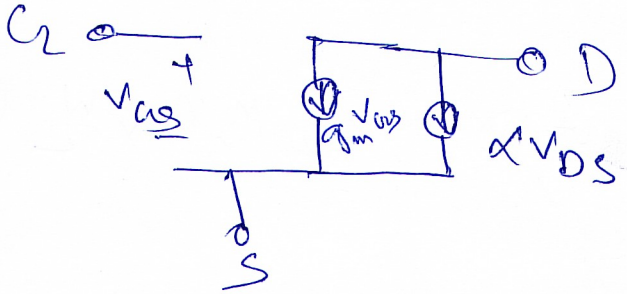
(c) deep triode region.

Small-Signal Model:

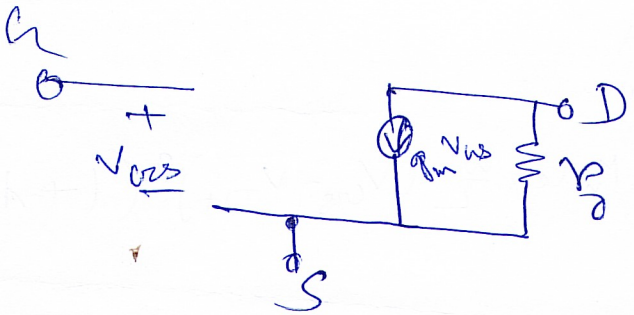
If the bias currents and voltages of MOSFET are only slightly disturbed by signals, the nonlinear, large-signal models can be reduced to linear, small-signal representations.



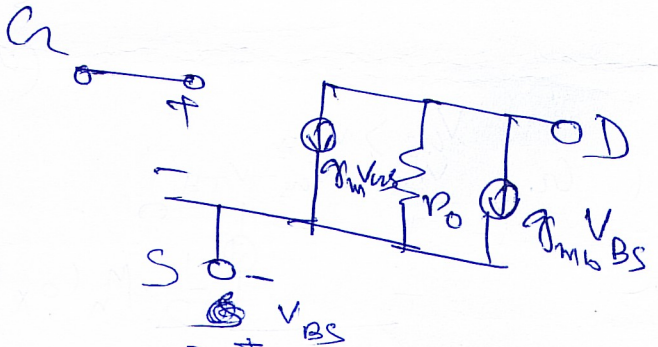
(a) Basic Small Signal model



(b) Channel-length modulation represented by a dependent current source



(c) Channel-length modulation represented by a resistor.



(d) Body effect represented by a dependent current source.

We derive the small signal model by producing a small increment in a bias point and calculating the resulting increment in other bias parameters. Since the drain current source equal to or 1/ (Figure)

Note that the low-frequency impedance between G and S is very high. This small-signal model of an ideal MOSFET.

Owing to channel-length modulation, the drain current also varies with the drain-source voltage. This effect can also be modeled by a voltage-dependent current source, but a current source whose value linearly depends on the voltage across it is equivalent to a linear resistor (Fig. c). Tied between D and S, the resistor is given by

$$r_o = \frac{\partial v_{DS}}{\partial I_D} = \Big/ \frac{\partial I_D}{\partial v_{DS}} = \frac{1}{\frac{1}{2}\mu_n C_{ox}\frac{W}{L}(v_{GS}-V_{TH})^2 \cdot \lambda}$$

$$\approx \frac{1}{\lambda I_D}$$

The output resistance, r_o, impacts the performance of many analog circuits. For example r_o limits the maximum voltage gain of most amplifiers.

ow recall that the bulk potential influences the threshold voltage and hence the gate source overdrive. With all other terminals held at a constant voltage, the drain rent is a function of the bulk voltage. That is, the ulk behaves as a second gate. Modeling the dependence y a current source connected between D and S (Fig. d), e write the value as $g_{mb}v_{bs}$, where $g_{mb} = \partial I_D/\partial v_{BS}$ n the saturation

$$g_{mb} = \frac{\partial I_D}{\partial v_{BS}}$$

~~FET-BJT~~

$$= \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH}) \left(- \frac{\partial V_{TH}}{\partial V_{BS}} \right)$$

we also have

$$\frac{\partial V_{TH}}{\partial V_{BS}} = - \frac{\partial V_{TH}}{\partial V_{SB}} = - \frac{\gamma}{2} (2\phi_F + V_{SB})^{-1/2}$$

Thus

$$g_{mb} = g_m \frac{\gamma}{2\sqrt{2\phi_F + V_{SB}}} \quad \text{---- (1)}$$

$$= \eta g_m.$$

where $\eta = \frac{g_{mb}}{g_m}$. As expected, g_{mb} is proportional to γ. Eqn (1) also suggests that incremental body effect becomes less pronounced as V_{SB} increases. Note that $g_m V_{GS}$ and $g_{mb} V_{BS}$ have the same polarity, i.e raising the gate voltage has the same effect as raising the bulk potential.

Problem: A MOSFET is biased at a drain current of 0.5 mA. If $\mu_n C_{ox} = 100 \mu A/V^2$, $\frac{W}{L} = 10$, and $\lambda = 0.1 V^{-1}$ calculate its small-signal parameters.

Ans. we have

$$g_m = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D} = \frac{1}{1 k\Omega}$$

Also

$$r_o = \frac{1}{\lambda I_D} = 20 k\Omega$$

This means that the intrinsic gain, $g_m r_o$ is equal to 20 for this choice of device dimensions and bias current.

H.W Repeat the above example iff W/L is doubled.?